

CMS Workflow Management Review 2025

August 4, 2025

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Introduction

A CMS workflow management system review was commissioned by the CMS Offline and Computing Level 1 managers in October of 2024.¹ The scope and charge are below:

Scope:

The workload management system of CMS covers 4+1 different L2 areas of CMS' O&C organization:

1. WMDM (Workflow Management and Data Management)
2. AIS - AKA CRAB and associated user support (Analysis)
3. Submission Infrastructure
4. Operations
5. In addition there are systems outside of O&C (eg. MCM) that participate in the interface between physics users and WM that should be examined.

Charge:

- Find out the level of satisfaction with using the system under evaluation; satisfaction with stability and error rates. If it is working well, find out why, is it hero mode? How much time is spent in development, and operations for people since we can objectively measure computer time.
- Getting operations buy-in is important, find out the cost in \$\$ to sustain the existing solution.
- For CMS, document the lessons learned from the Rucio transition. It was a large-scale transition from one solution to another and should inform any similar move in workload management.
- Determine: Are newly developed workflow languages useful to us? or is that only for the analysis use case (DASK or others) <https://indico.cern.ch/event/1380367/timetable/?view=standard>, some evaluation of this question would be very useful to document.

¹  Workload Management Review Charge 2

- Define a roadmap towards a sustainable workflow management system for CMS.

The last time these components were reviewed was in 2018².

Changes Since the Last Review

The previous review centered on three key questions:

- Does CMS have a feasible plan to address the requirements of the HL-LHC?
- To what extent does it make sense to maintain separate workflow management tools for production and analysis? Could unifying these tools reduce technical or human resource costs?
- Could tools developed outside CMS be integrated into the workflow management infrastructure to meet new requirements?

While the scope of the current review aligns closely with that of the 2018 review, the context has changed significantly over the intervening seven years. With the HL-LHC now only five years away, the focus has shifted from assessing the feasibility of plans to evaluate whether CMS is truly on track to be ready.

The 2018 review classified system components into three categories: those requiring minimal changes, those that could scale horizontally, and those in need of structural redesign. There were only a few components identified as needing structural changes. The scope of the 2018 review was focused on technical capability and was not asked to examine long-term sustainability. For example, PhedEx was deemed sufficient from a scalability standpoint, yet it had to be decommissioned shortly thereafter due to the lack of sustained development effort.

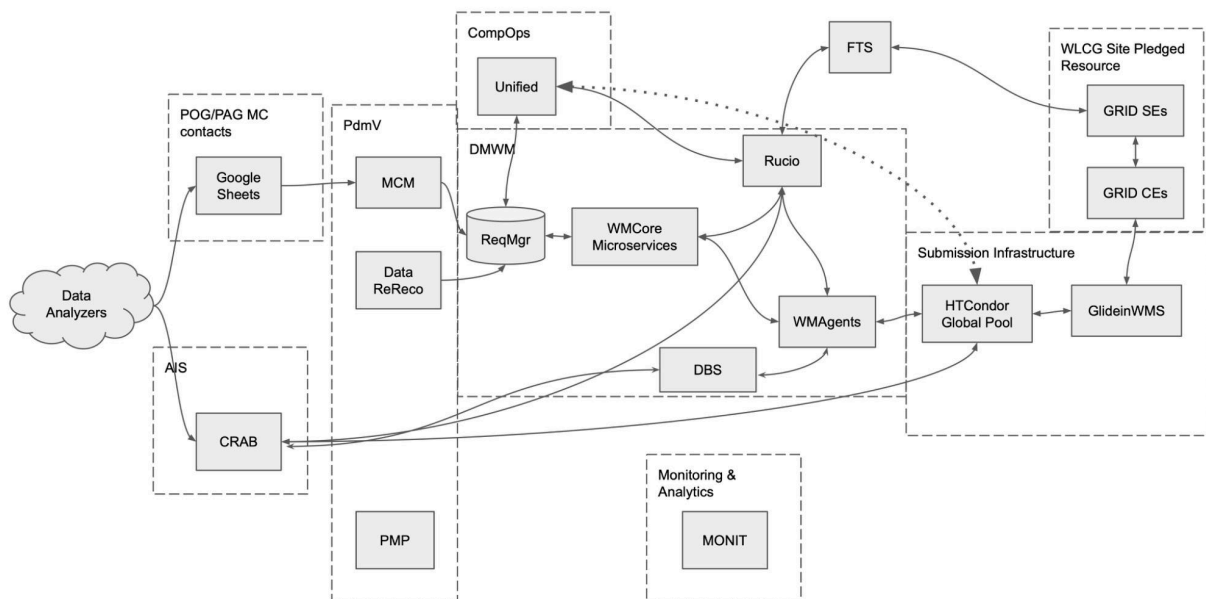
At that time, the review recommended aligning the user analysis system more closely with production workflows by adopting shared workflow components, while allowing for a distinct user interface. This recommendation was not implemented, however there are some shared modules between the two systems. As a result, the current review begins from the observation that there is **no established plan** for user analysis or user-defined simulation workflows on WLCG infrastructure for the HL-LHC era.

Finally, the 2018 review considered whether incorporating external tools into the CMS workflow system would be beneficial and concluded that doing so was not necessary. In contrast, the current review is explicitly tasked with evaluating whether entire external workflow systems could replace parts of the CMS infrastructure—and whether doing so could reduce development effort, improve sustainability, and enhance maintainability.

In general, the 2018 review was optimistic about the scalability of the system in preparation for the HL-LHC. While the fundamental challenges posed by the HL-LHC remain unchanged, the context in which they must be addressed has evolved considerably. The workflow components under review are now eight years older, and the pool of available experts and developers has diminished. At the same time, the distributed computing environment has grown more diverse and complex, with increased reliance on HPC resources and soon a growing number of workflows designed to leverage accelerated computing architectures such as GPUs are expected. These shifts raise additional questions about the long-term viability, adaptability, and sustainability of the current system architecture.

The Review Process

The review process began in October of 2024. The review committee was chosen from computing experts primarily within CMS, but included an external expert as well. The committee decided to begin with a detailed series of discussion of the 6 primary areas as described in Figure 1 individually.



Original Diagram created by Nick Smith

Figure 1: Box diagram of the CMS workflow management system

Each area was asked a standardized set of questions.

- Can you describe the architecture of your solution? What are its components and how do they interact?
- How many developer FTEs are supporting your solution?
- How many people are involved in operating the system and documenting it?

- Where does funding / person power for the solution come from?
- What are the evolutionary plans for this component?
- Is your solution used by other organizations and does it constrain evolution?
- What is going well and what functionality is missing?
- What progress have you made in testing the scalability of your solution to Run4 requirements?

The discussions began with the “Submission Infrastructure” and “Monitoring and Analytics” and moved to “DMWM Core” and “CompOps”. The final 2 components were “CRAB” and “PdmV”. Once the CMS components had been investigated other workflow management solutions were asked to present also following the defined questions. The two systems explored were PanDa, which originated in ATLAS, and DIRAC, from LHCb. A total of 20 review meeting sessions were convened, including two face-to-face meetings.

Overall CMS Workflow Management Team Structure

The CMS Workflow Management system is organized as a set of cooperating Level-2 tasks. Oversight is provided by the Level-1 Coordinators for Offline and Computing, who serve as the primary aggregation point for task management. However, there is no designated individual or team responsible for overall architectural decisions across the system. The Level-1 Coordinators rotate on a time scale that does not let them substitute as cross system architects.

After more than 15 years of operation, some development choices have been driven by immediate needs rather than long-term sustainability. In several areas, the boundary between development and operations is fluid—operations teams have taken the lead on substantial development efforts to extend or enhance the workflow management system’s functionality.

Pain Points and Limitations in the Current System

Lack of available experts in the core
Need to add additional layers for operations tasks
Long tails to complete and release production workflows

Despite a long history of enabling largely successful processing of CMS data and generation of MC, there are undeniable limitations in the current system:

- *Because of the organic way that the software has evolved, it has become difficult to evolve further. Changes must be made carefully and with expert-level knowledge to avoid introducing bugs in other parts of the system. As a result, only a limited group of seasoned experts can maintain and adapt the software, limiting the available effort to help with development and even sometimes operational tasks.*
- *Through the history of CMS computing, as operational experience grew and operational procedures evolved, it became clear that not all desired operational features were present. Perhaps unsurprisingly, these gaps were initially filled through software developed on an ad-hoc basis by the operations team. Unfortunately, this functionality*

has not been subsequently well integrated into the core software, necessitating the continued maintenance of auxiliary code by the operations team to supply the missing functionality.

- *Historically, there have been concerns about the presence of long tails in the processing of workflows. A variety of contributing factors have been explored, including reliability of the WM components, blind spots in monitoring, gaps in operational automation, and flaws in CMS policies around managing and prioritizing work. In addition CMS moved to a different data management solution long after (2018) the original design of the workload management system (2010) leading to insufficient integration of the two systems WM and DM. Although a definitive solution to these concerns has not surfaced, it is clear that the complexity of the current code base, and the inability of people beyond the seasoned experts to diagnose issues, presents a significant obstacle in reducing processing tails.*
- *Many of the required properties of datasets; naming, extensibility, and internal structure were realized after the initial design and are not easily produced in the current system.*

Assessment of Sustainability

Many components of the CMS Workflow Management system were originally developed over 15 years ago. To meet evolving requirements, additional operational layers have been built around these core elements. While this layered approach has enabled CMS to continue fulfilling its scientific mission, it has also introduced a form of gridlock: deficiencies in the foundational components cannot be addressed without disrupting the overlaying services.

The development of both the core and operational layers has proceeded without a unified management or coherent evolution plan. Today, only a handful of developers have a deep understanding of the core system, and no team exists with comprehensive knowledge of the full software stack. The external services on which the Workflow Management and Data Management (WMDM) components depend are still maintained, but they reflect past decisions and are not aligned with the choices one would make in designing a modern system from the ground up.

The developers most familiar with the current core architecture believe it cannot be efficiently modernized. A complete redesign—either from scratch or by leveraging elements from an external, shared project—is likely the most viable path forward. While it's difficult to declare any software system end-of-life: two parallel development tracks have been evolving independently for years; the expertise needed to upgrade the legacy components in place is lacking. We could try to reconstitute that expertise and continue to evolve the system, but it would most likely require more effort and would be constrained by some of the original choices. It seems most efficient to learn from the successes, lessons learned and required adjustments of the original workflow management system and have a fresh start with modern tools and techniques.

Workflow Elements Definition

Traditional HTC Processing

This category includes event reconstruction, event simulation, and data reduction tasks such as skimming, slimming, and filtering and heretofore has represented the vast majority of computing resource consumption. Workflows may be centrally submitted or initiated by users, and can include inference or application of AI/ML algorithms. Traditional high-throughput computing (HTC) assumes that resource orchestration is managed externally. These workflows are typically characterized by predictable, static resource allocations and may execute on WLCG sites, cloud platforms, or HPC resources.

End User Analysis Processing

Covers interactive or semi-interactive workflows run from the output of the centralized processing and production campaigns. These workflows may prioritize responsiveness, interactivity, flexibility, and integration with user-defined software environments and may run at dedicated analysis facilities, providing streamlined access to data and computing resources for end users.

Tightly Coupled Processing

Refers to workflows that require strong inter-process communication and coordination, such as MPI-based applications and AI/ML model training. These workflows typically manage their own resource orchestration along with the local scheduling system and are optimized for execution on tightly integrated computing environments, including HPC systems with specialized interconnects or accelerators.

Specialized Capabilities of the CMS Workflow and Data Systems

Over time, CMS has developed several capabilities that provide flexibility, scalability, and operational efficiency in its workflow and data management systems. As plans to evolve the system are considered, it is critical that these core strengths are preserved—or that any proposed changes are backed by a compelling technical rationale.

- **Late Binding and Pilot-Driven Resource Scheduling:** CMS was an early adopter of pilot-based late-binding resource scheduling at the LHC. This approach significantly simplifies resource discovery and planning by ensuring that resource information is both accurate and current at the time of job execution. The pilot model has enabled CMS to implement whole-node scheduling and support multi-task pilots, offering improved efficiency and utilization. It is particularly well-suited for increasingly heterogeneous environments, as pilots dynamically assess available resources—including CPU and accelerator configurations—at runtime.
- **Data Federation:** The CMS data federation spans the entire distributed computing environment, enabling flexible and efficient data access. CMS has developed data structures optimized for remote access, including aggressive read-ahead strategies that

allow workflows to stream data over wide-area networks without significant performance degradation. This capability has decoupled processing from strict storage locality, enabling broader site participation and empowering Tier-3 centers and even individual researchers to contribute effectively using local resources.

- **Work Chaining:** CMS has the ability to chain multipart workflows together. A multipart workflow is one in which the computation either must be or can benefit from being split into multiple cmsRun invocations, with subsequent cmsRun invocations processing the outputs of prior ones. CMS currently supports two modes of chaining: *Task Chains* in which each part of the multipart workflow is implemented in a different batch job, with the outputs staged out between tasks, and *Step Chains* in which all parts are executed as part of a single batch job, with outputs remaining in the local storage of the worker node executing the batch job. Task Chains are generally more flexible, allowing different parts of the multipart workflow to have different resource requirements and work granularity, but also tend to be logistically more challenging, requiring staging out intermediate datasets and making it harder to monitor workflow progress. In addition task chains are more vulnerable to stage out and site issues, leading to longer tails. The operations group prefers to minimize the number of workflows run this way and has developed scripts that convert task chains to step chains. Step Chains, while generally simpler to work with and more lightweight, especially in terms of I/O, are not easily able to adapt for workflows where the multiple parts require different resources, which can result in poor resource utilization. Both aspects of work chaining have their uses, and CMS should retain—or even enhance—its ability to support chaining of multipart workflows, a design that takes advantage of the best of both types is highly desired.

High Level Assessment of external possible solutions

Workflow management is a shared challenge across all high-energy physics (HEP) experiments, and several mature external solutions exist. As part of the review, the committee interviewed representatives from two comparable systems: PanDA, developed by the ATLAS collaboration, and DIRAC, used primarily by LHCb.

Both tools have extended their reach beyond their original experiments. PanDA is also used by the Vera C. Rubin Observatory and AMS, while DIRAC supports Belle II, ILC, and CTA. These external adoptions underscore the common aspects of workflow management across programs.

The committee focused on several key aspects:

- Sustainability and size of the support and development teams
- Involvement of external user communities
- A qualitative assessment of user satisfaction

Observations:

- Both PanDA and DIRAC benefit from sizable support teams. In the case of PanDA, the team is dramatically larger than the equivalent CMS team by a factor of 3 to 4. DIRAC has a team that is comparable in size to CMS.
- Each project has mechanisms to engage contributors from outside their originating experiment. However, the structure and maturity of these mechanisms differ.
 - DIRAC is organized as a formal collaboration, with regular collaboration meetings, structured membership, and coordinated software releases.
 - PanDA has been adopted successfully beyond ATLAS, but it is less clear how external development is coordinated. Integration of non-ATLAS contributions appears to be more ad hoc, and ATLAS uses the latest (head) release in production, which may complicate external adoption.
- PanDa also uses a pilot approach for work submission, so it would be potentially easier to maintain the CMS core competencies late binding by collaborating with PanDa.

Overall, both tools demonstrate the viability of shared workflow solutions across scientific domains, but differ in their governance models and approaches to sustainability and external engagement.

Key Finding

CMS computing has successfully supported the full physics program from the preparation phase through Run 3, enabled by innovative development and specialized capabilities. However, as functional and scale requirements have grown, workflow management has evolved reactively—driven by operational needs rather than a coordinated, sustainable plan. The committee finds that CMS does not currently have a workflow management system suitable for Run 4 and beyond, nor a clear plan to develop or adopt one. The existing system is assessed by its experts as unsustainable and not evolvable. A viable path forward requires CMS to commit and sustain a multi-year effort to build or integrate a new system.

Recommendations

The committee recommends that CMS develop a resource-loaded plan for a sustainable workflow management system that meets the needs of Run 4 and remains viable through the end of the experiment after Run 5. Each of the phases will need an executor to be responsible for completing the task and documenting the deliverable. The following phased steps are proposed to achieve this objective:

Step 0 (September – November 2025): Requirements and Planning

- CMS should formally define the requirements for a future workflow management system, addressing both traditional HTC workflows and the potential need for tightly coupled computations.
- CMS should identify the elements needed to support user defined workflows and commit to designing a single system to support both production and user defined workflows.
- The collaboration must start to identify available and sustainable effort for a two-year development and integration project, targeting 4–5 FTEs starting in summer 2026. Options for securing this effort should include institutional commitments, national contributions, and external contracts if needed.
- To enable this transition, CMS should implement a feature freeze across the entire workflow management system. This freeze, already underway for CRAB and user analysis, should be expanded and formally assessed. During the six-month Step 0 period, CMS should compile a prioritized list of critical tasks necessary to support Run 3 data collection and processing, and define a clear plan and endpoint to complete them. The feature freeze is discussed in the appendix of this document.
- The gathering of requirements could be facilitated by a dedicated workshop bringing together the relevant stakeholders.

Step 1 (November – March 2026): Blueprint Development

- Review level 2 organization and define a more optimal cross cutting project structure
- CMS should document a decision regarding whether to build a new system or adopt an existing solution based on identified effort, capabilities, guidance of funding agencies and potentially prototype investigations.
- Based on the requirements and available resources identified in Step 0, CMS should develop a blueprint for the new/adopted workflow management system. This blueprint must:
 - Preserve the unique technical capabilities of CMS,
 - Specify the overall software architecture,
 - Provide time and effort estimates.
- It is essential that the blueprint development be undertaken by a team that includes both experienced developers and operators of the current system as well as new contributors with fresh perspectives and fewer preconceptions.
- It would be desirable to involve a set of advisors with a broad perspective beyond workflow management to provide regular feedback, for example via monthly meetings, to ensure design efforts stay aligned with requirements.
- The blueprint should be presented to the CMS O&C Level-1 Coordinators and forwarded to the CMS Management Board for approval.

Step 2 (April 2026 – July 2028): Implementation

- Embarking on this step is contingent on effort

- CMS should execute the plan outlined in Step 1 using the resources secured in Step 0.
- This two-year development effort should support:
 - System implementation,
 - Testing during preparatory data challenges,
 - Deployment of a robust, production-ready service by the start of Run 4.
- During this effort, it is important that the O&C L1 managers and advisory team from the blueprinting effort remain engaged and available to provide feedback, with at least quarterly updates.
- O&C L1 managers should keep the implementation team apprised of any LHC schedule changes and their impact on the development timeline.
- CMS should remain vigilant against the possibility of development effort being drained away by issues arising with the current WM system.

Outlook

CMS has achieved remarkable success with its current workflow management system, supporting the full physics program over the past 15 years and enabling the publication of over a thousand scientific papers. However, the system has evolved reactively, with operational patches layered onto an aging core architecture. While this approach has sustained operations through Run 3, it is no longer viable for the challenges posed by Run 4 and the HL-LHC era.

The committee finds that CMS is now at an inflection point. The current system is not sustainable, and no comprehensive plan exists to develop or adopt a replacement. The core expertise needed to maintain or evolve the legacy system has eroded, and rebuilding that capacity would require a commitment comparable to designing a new system.

To ensure readiness for Run 4 and long-term viability through the end of the experiment, CMS must act decisively. A feature freeze, clear requirements definition, and sustained development effort are needed to support the design and implementation of a modern, maintainable workflow system. This system must preserve the specialized capabilities that have contributed to CMS's success—such as pilot scheduling, data federation, and workflow chaining—while addressing new demands including heterogeneous computing environments and user-defined workflows.

It is important to note that the recommendations provided above depend critically on identifying the necessary effort. Should it prove impossible to identify the appropriate effort, the recommended plan will not be feasible. In that case, the only alternative to move forward with the current effort level would be to substantially descope the plans, either to try to make minimal alterations to the existing codebase to patch only the most critical deficiencies, or to move to adopt another software solution (e.g. Panda or DIRAC) wholesale, making no attempt to adapt that software to preserve the unique CMS capabilities provided by the current system. It is likely that some requirements may not be met at all in this scenario and CMS could lose its specialized capabilities. It is the determination of this committee that such a scenario carries an

untenable risk to CMS's HL-LHC physics program, and therefore we do not recommend it as a viable alternative.

With the HL-LHC fast approaching, the window to act is narrow. CMS must commit now to a structured, multi-phase roadmap to secure the computing infrastructure needed for the next decade of discovery.

Appendices

Impact of a Feature Freeze

Currently, the operation of the CMS workflow management system relies heavily on the involvement of the core development teams. It is estimated that up to 50% of the developers' time is dedicated to devops and operational support activities. This level of ongoing support significantly limits the capacity available for system evolution and strategic development. To successfully develop or transition to a new workflow management system within the expected timeframe, it will be essential to substantially reduce the support burden for the existing system. Moving forward, only critical issues that directly impact data processing or halt production should be addressed. Enhancements, quality-of-life improvements, and other non-critical feature requests will need to be deferred.

This implies a de facto feature freeze on the current system, beginning in 2026 and lasting two to three years. While this freeze is necessary to enable focused development of the next-generation system, it will have a tangible impact on user experience and operational flexibility. As a specific example, CMS will be unable to implement run dependent MC for Run3. The community must be prepared for a period of limited functionality growth, during which efforts will be concentrated on sustaining essential operations while redirecting resources to future infrastructure. Clear communication and prioritization will be required to manage expectations and ensure that the most urgent needs continue to be met.

Discussion of the CMS Adoption of Rucio

The CMS adoption of Rucio for data management serves as a valuable case study in the successful integration of an external tool originally developed by a competing experiment. It offers useful parallels to the current deliberations around the future of CMS workflow management.

One of the reasons Rucio was a good fit is that data management has a narrower and more uniform scope across the LHC experiments compared to workflow management. All experiments store data as files optimized for transfer and archiving, with files generally immutable after creation, globally uniquely identified, and logically organized into datasets. These commonalities made it feasible for Rucio to serve as a near drop-in replacement for

CMS's previous system, PhEDEx—though this should not understate the considerable effort involved in integration.

Workflow management, by contrast, varies more significantly between experiments. Each collaboration tailors its systems to specific needs and derives competitive advantages from that specialization. Consequently, adopting a common solution such as DIRAC or PanDA may require substantial development to support CMS-specific dependencies and operational models.

To its credit, Rucio was already evolving into a community-oriented, multi-experiment tool before CMS formally adopted it. This proactive design choice fostered a sense of shared ownership and encouraged contributions from diverse communities—an essential ingredient for long-term sustainability. If CMS chooses to adopt an external workflow management system, it will be critical to select a platform where CMS can have a meaningful voice in shaping its direction and priorities.

Another factor that smoothed CMS's transition to Rucio was the lack of a viable in-house alternative. With the PhEDEx development team no longer available, CMS would have needed to assemble a new team to build and maintain a standalone solution. Rucio offered a technically sound and maintainable option that aligned with the project's resource realities.

The experience with Rucio illustrates that adopting external tools—particularly those designed with multi-experiment collaboration in mind—can be both pragmatic and beneficial. CMS should remain open to similar strategies for workflow management, even when internal alternatives exist, provided that the chosen solution meets technical requirements and offers opportunities for influence and collaboration.